(a) Frame camera → RGB encoder



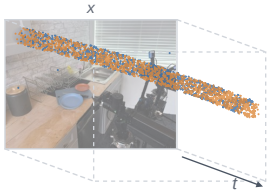
RGB frames at 30 fps (motion blur)

every frame, all H×W×3 values

ResNet / DINOv2
dense MACs

33 ms resolution, blur
compute ∝ pixels × frame rate

(b) DVS → accumulate events → fuse with RGB → RGB encoder



DVS events (x, y, t, p), µs timestamps

sum over ΔT



event frame



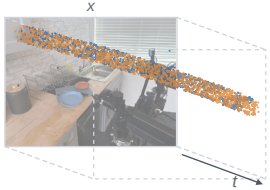
RGB frame

concat

ResNet / DINOv2
dense MACs, zeros included

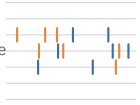
events re-binned into frames
sparsity and µs timing lost

(c) DVS → spiking neural network: purely event-driven



same event stream, no frames

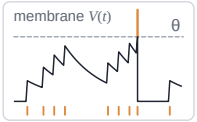
event = spike



spike trains
(active pixels only)



SNN (LIF neurons)
only spiking neurons update



integrate → fire → reset

compute only on spikes
static scene → ≈ no compute